CS-477 COMPUTER VISION

SEECS CV Face-Off 2026

Semester Project Guide

Research -> Prototype -> Integrate -> Deploy -> Validate -> Face-Off

Target Platform: NVIDIA Jetson Orin Nano 8GB

14-Lab semester execution model | 2-student technical sub-groups | Progressive integration

1. Project Overview and Expected End Goals

The semester project is a continuous systems-engineering exercise rather than a final-week demonstration. Each project team is divided into small technical sub-groups that develop independently against agreed interfaces, then progressively integrate their modules into a complete embedded computer-vision system.

Project 1 - Automatic Class Attendance Monitoring

Develop an embedded classroom vision system capable of detecting students, assessing face usability, identifying enrolled students, maintaining identity across video, and converting repeated visual evidence into a reliable attendance record.

- End-to-end target: Camera -> Face Detection -> Quality/Alignment -> Recognition -> Tracking -> Attendance Decision -> Attendance Record.
- The final solution must distinguish enrolled students from unknown persons and avoid marking attendance from isolated or unreliable observations.
- The complete pipeline must demonstrate deployability on the Jetson Orin Nano 8GB with measured accuracy, latency/FPS and robustness.

Project 2 - Driver Drowsiness Monitoring

Develop an embedded driver-monitoring system that observes facial and behavioral cues over time and produces a robust drowsiness state and alert, rather than relying on a single-frame decision.

- End-to-end target: Camera -> Face/Landmarks -> Eye/Facial Cues -> Temporal Analysis -> Drowsiness Decision -> Alert.
- The system should combine complementary evidence such as eye closure/blink behavior, yawning/facial cues and persistence over time.
- The complete pipeline must demonstrate deployability on the Jetson Orin Nano 8GB with measured detection/decision performance and real-time behavior.

A high-performing PC prototype is not the final goal. The end goal is a measurable, integrated and deployable embedded CV system.

2. Division, Scope and Justification of Technical Sub-Groups

Sub-groups are intended to be approximately two students each. Each owns a self-contained technical challenge with explicit inputs, outputs, evaluation metrics and an integration responsibility. This allows concurrent development and prevents one unfinished upstream module from stopping the entire team.

2.1 Project 1 - Attendance Monitoring

SG	Technical Challenge	Interface	Scope	Why Separate?
SG-1	Classroom Face Detection	Frame -> face boxes + confidence	Reliable multi-face/small-face localization under classroom scale, pose, lighting and occlusion.	Creates a measurable front-end and isolates detection performance from recognition.
SG-2	Face Quality, Alignment & Preprocessing	Face ROI -> quality/accept-reject + normalized face	Assess blur/size/pose/illumination; align and normalize usable faces.	Separates image-quality failure from identity-recognition failure.
SG-3	Student Face Recognition	Normalized face -> Student ID/UNKNOWN + confidence	Enrollment, identity matching, thresholding and open-set rejection.	Makes identification independently measurable and prevents unknown persons being forced into known classes.
SG-4	Multi-Person Tracking & Identity Association	Detections/identity evidence -> persistent Track ID + identity	Maintain continuity, reduce repeated recognition and control ID switches/re-entry.	Adds temporal consistency and supports efficient embedded operation.
SG-5	Attendance Decision & Validation	Track/ID/time/confidence -> attendance decision + evidence	Fuse repeated observations into Present/Absent decisions; control false and missed attendance.	Teaches that recognition accuracy is not the same as system-level attendance accuracy.
SG-6	Embedded Deployment & System Integration	Modules -> integrated Jetson application	Camera, environment, module integration, profiling, optimization and end-to-end execution.	Makes embedded constraints a design input throughout the semester rather than a final-week porting task.

2.2 Project 2 - Driver Drowsiness Monitoring

SG	Technical Challenge	Interface	Scope	Why Separate?
SG-1	Driver Face & Landmark Detection	Frame -> face/landmarks + ROIs	Reliable face localization and relevant eye/mouth regions under realistic driver pose and lighting.	Provides a stable geometric front-end for all downstream cue analysis.
SG-2	Eye State & Blink Analysis	Eye ROI/landmarks -> eye state/closure evidence	Detect eye openness/closure, blink behavior and sustained closure using suitable visual features/models.	Eye behavior is a major but independently testable drowsiness cue.
SG-3	Yawn & Facial-Cue Analysis	Face/mouth ROI -> yawn/fatigue cues + confidence	Detect yawning and complementary facial fatigue cues.	Provides evidence independent of eye closure and improves robustness to single-cue failure.
SG-4	Temporal Behaviour Analysis	Frame-level cues -> persistent temporal indicators	Model duration, frequency and temporal persistence rather than isolated frame classifications.	Drowsiness is inherently temporal; this module converts observations into behavior.
SG-5	Drowsiness Decision & Alert Logic	Temporal cues -> drowsiness state + alert	Fuse cues, set decision thresholds/states and minimize nuisance/false alerts.	Separates cue detection from the final safety decision and permits system-level evaluation.
SG-6	Embedded Deployment & System Integration	Modules -> integrated Jetson application	Camera, environment, integration, profiling, optimization and real-time alert execution.	Ensures the complete system is engineered for the common embedded target.

3. Project Execution Guidelines

3.1 Concurrent Development and Interface-First Design

- All sub-groups begin concurrently. No pair should wait for the preceding module to be completed.
- By the end **of Lab 4**,
  (a) one group meeting and discussion on overall scope of project;
  (b) finalize the sub-group level diagram of the project;
  (c) each module must have a frozen V1 input/output contract: data format, units/coordinate convention, confidence representation, error/empty-output behavior and example test data.
  (d) selection of the group leader (one for whole semester or one per month)
  (e) github project setup (repository & project board)
- Each downstream group must be able to develop against either the real upstream output or standardized mock input.
- Internal algorithms may change without affecting other groups, provided the agreed interface remains compatible. Interface changes after Lab 4require team-level agreement and documentation.
- Integration is progressive: interface testing first, multi-module integration next, then complete Jetson deployment.

3.2 Git/GitHub Working Practice

- One repository per major project team; each technical module has a clearly owned folder.
- Use a protected/stable main branch for integrated code. Sub-groups work on feature/module branches and merge through reviewed pull requests.
- Commit frequently with meaningful messages that describe the technical change; commit count alone is not a grading metric.
- Every module folder should contain a short README: purpose, dependencies, input/output interface, how to run, test data and current performance.
- Do not commit large raw datasets, trained-model caches or environment folders unless specifically approved; maintain reproducible download/setup instructions instead.
- Tag or preserve working versions at formal integration gates so a failed later change does not destroy the last known-good system.

3.3 Recommended Repository Structure

```
project/
  interfaces/        # schemas, examples, mock inputs/outputs
  datasets/          # manifests, labels, acquisition/preparation scripts
  module_sg1/
  module_sg2/
  module_sg3/
  module_sg4/
  module_sg5/
  embedded_sg6/
  integration/
  evaluation/
  documentation/
```

3.4 Inter-Subgroup Coordination

- Nominate one Technical Team Lead/Coordinator for system-level coordination; this role does not remove responsibility from individual sub-groups.
- Hold a short weekly technical sync focused on interface status, blockers, integration readiness and Jetson constraints.
- Maintain a simple module status: GREEN = on track, AMBER = at risk, RED = blocking integration.
- A blocking issue must be escalated early. Teams should not silently compensate for an inactive pair until the final integration phase.
- SG-6 owns the common Jetson environment and integration framework, but every algorithm sub-group remains responsible for making its module deployable and benchmarkable on the target platform.

- At demonstrations, either student in a pair may be asked to explain the algorithm, experiment, interface or code contribution.

3.5 Standard Experimental Discipline

1. Define the technical question.
2. Research 2-3 candidate approaches where practical.
3. Implement a baseline.
4. Run controlled experiments and tune parameters.
5. Measure with agreed quantitative metrics.
6. Compare accuracy/robustness against computational cost.
7. Select the method using evidence, not convenience.

Every major checkpoint should answer: What did you try? What did you measure? What did you learn? What engineering decision follows?

4. Assessment Criteria

Assessment is continuous. A strong final demonstration cannot erase missing semester work, and a student is not automatically entitled to the full team score because the integrated system performs well.

The semester project will be assessed at **two levels**:

1. **Sub-Group Performance** — performance of each 2-student technical sub-group throughout the semester.
2. **Overall Project Performance** — performance of the complete integrated system developed by the full project team.

A separate **SEECS CV Face-Off bonus** will be awarded to the winning teams.

Project Grade

Project Grade = 60% Sub-Group Performance + 40% Overall Project Performance + Face-Off Bonus

The normal project grade is out of **100%**. The Face-Off bonus can improve the grade up to a maximum of **100%**.

4.1 Sub-Group Performance — 60%

Each 2-student sub-group will be assessed continuously through its lab milestones, technical deliverables, experiments and contribution to integration.

Criterion	Weight within Sub-Group	Contribution to Project Grade
Problem Understanding & Research	15%	9%
Algorithm / Prototype Quality	25%	15%
Experimentation & Quantitative Evaluation	25%	15%
Interface Compliance & Reproducibility	15%	9%
Jetson Readiness / Performance	10%	6%
Individual Contribution & Technical Understanding	10%	6%
Total	**100%**	**60%**

The sub-group assessment will consider **continuous progress**, rather than only the final implementation. Students should be able to demonstrate:

- understanding of the technical problem and candidate approaches;
- a working and technically appropriate implementation;
- quantitative experiments, comparisons and failure analysis;

- compliance with agreed inter-module interfaces;
- readiness for Jetson deployment and integration; and
- individual understanding and contribution to the work.

Both students in a sub-group are expected to understand their complete module. **Individual marks may differ** where there is clear evidence of unequal contribution or technical understanding.

4.2 Overall Integrated Project Performance — 40%

The complete 10–12 student project team will also receive a common system-level assessment.

Overall Project Criterion	Contribution to Project Grade
End-to-End Functional Completion	15%
System-Level Accuracy & Robustness	10%
Integration Quality	5%
Embedded Jetson Performance	5%
Final Validation & Demonstration	5%
Total	**40%**

Particular emphasis will be placed on **completion of the end-to-end system**. Strong individual modules alone do not constitute a completed project.

Project Completion Levels

Final System Status	Maximum Overall Project Score
Complete: Functional end-to-end system demonstrated on Jetson	**40/40**
Partially Integrated: Major pipeline operational but an essential function/module remains incomplete	**Maximum 30/40**
Modules Only: Individual modules demonstrated without a functional end-to-end system	**Maximum 20/40**

This ensures that every sub-group has an incentive to support **system integration and overall project completion**, in addition to completing its own technical work.

4.3 SEECS CV Face-Off 2026 Bonus

The final integrated systems will compete under common evaluation conditions in the **SEECS CV Face-Off 2026**.

The winning team in each project category will receive:

+5% Project Grade Bonus

The bonus is added **after** the normal project grade is calculated, subject to a maximum final Project Grade of **100%**.

For example:

Sub-Group Score = 82%
Overall Project Score = 90%

Project Grade:

(0.60 × 82) + (0.40 × 90) = 85.2%

If the team wins the Face-Off:

Final Project Grade = 85.2 + 5 = 90.2%

The Face-Off therefore provides an additional incentive for excellence without reducing the academic grade of a technically successful team that does not win the competition.

4.4 Individual Accountability

Although the project is team-based, individual contribution remains assessable throughout the semester. Evidence may include:

- Git/GitHub contributions;
- participation in lab milestones and demonstrations;
- implementation and experimental work;
- contribution during integration;
- individual technical questioning; and
- ability to explain the sub-group's algorithms, results and design decisions.

5. 14-Lab Execution Roadmap and Integration Gates

Lab	Focus	Minimum Project Deliverable
1	Problem & System Definition	Technical problem, system role, initial Input -> Processing -> Output, data needs and dependencies.
2	Research & Candidate Algorithms	Short literature/technology survey; 2-3 candidate approaches; limitations; baseline proposal; initial Jetson suitability.
3	Interface & Experiment Design	Frozen V1 interface, mock/test inputs, dataset/test samples, metrics and experimental plan. PHASE-I EXIT GATE.
4	Baseline Prototype	First working algorithm against standardized input; reproducible baseline result.
5	Alternative/Parameter Experiments	Second approach or meaningful parameter study; controlled comparison.
6	Quantitative Comparison	Metrics table, analysis and evidence-based method selection.
7	Improved Module & Robustness	Selected method improved and tested against difficult/failure cases.
8	**INTEGRATION GATE 1**	Replace mock data with neighboring real modules; first multi-module pipeline on development platform.
9	Jetson Module Deployment	Each module runs in target Jetson environment; camera/platform integration active.
10	Jetson Profiling & Optimization	FPS, latency, memory/compute profiling; optimization decisions documented.
11	**INTEGRATION GATE 2**	Complete camera-to-decision pipeline operating through target embedded environment.
12	End-to-End Optimization	System bottlenecks, thresholds, scheduling and reliability refined on Jetson.
13	Controlled/Blind Validation	System-level evaluation using held-out or controlled conditions; final failure analysis.
14	SEECS CV Face-Off 2026	Final embedded demonstration and comparative evaluation under common conditions.

Formal Integration Gates

- Lab 3 - Interface Gate: Mock Input -> Module -> Correct Interface Output.
- Lab 8 - Integration Gate 1: Real neighboring modules replace mock interfaces; interoperability is demonstrated.
- Lab 11 - Integration Gate 2: Complete camera-to-decision pipeline is operational on the target embedded environment.
- Labs 12-14 - Optimize and validate an existing integrated system; do not begin first-time integration here.

14 Labs -> Continuous Evidence -> Progressive Integration -> One Deployable Embedded CV System

Project Governance Principle

Sub-groups are accountable for their own technical deliverables, but the major project team remains collectively responsible for the integrated system. Early reporting of technical or participation risk is expected. A module that works in isolation but cannot integrate through its agreed interface is not complete.